ALEX SHENG

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EDUCATION

University of Michigan Aug. 2024 – Present
Ph.D. in Mathematics ◇ Advisor: Daniel B. Forger

University of Chicago Sept. 2020 – June 2024
B.A. in Mathematics with Honors

RESEARCH AND EXPOSITORY WRITINGS

Mathematics of AI ◇ Mathematical Biology ◇ Computational Neuroscience

- *Modeling Sleep Dynamics for Long-Term Psychiatric Risk Prediction.* Research work. In preparation
- *Performed Bach Trio Sonatas Reveal Topological Encoding of Polyphonic Music.* Research work. In preparation
- *From the topology of natural images to Topological CNNs.* Expository talk. PDF Apr. 2026

Number Theory ◇ Homotopy Theory ◇ Representation Theory

- *The S-unit equation using Lawrence-Venkatesh.* Expository writing. PDF Oct. 2025
- *Kolyvagin systems.* Expository writing. PDF Oct. 2025
- *Cofreeness of Lubin–Tate theory.* Expository writing. PDF Aug. 2023
- *Power operations and HKR character theory.* Expository writing. PDF June 2023
- *Elements of complex K-theory.* Expository writing. PDF Dec. 2022
- *Complex multiplication of elliptic curves and abelian varieties.* Expository writing. PDF Aug. 2022
- *On realizing rational and polynomial cohomology rings.* Expository writing. PDF Aug. 2021

CONFERENCES AND PROGRAMS

Arizona Winter School: Computational Aspects of Arithmetic Geometry and Cryptography Mar. 2026
IWoAT Summer School: Operads, Spectra, and Multiplicative Structures Aug. 2023
Park City Mathematics Institute Undergraduate Summer School: Quantum Computation July 2023
Preliminary Arizona Winter School: Heights in Diophantine Geometry Nov. 2022 – Dec. 2022
Midwest Algebraic Geometry Graduate Conference Aug. 2022

SELECTED COURSEWORK

At the University of Michigan

- EECS 553 Machine Learning
- MATH 564 Topics in Mathematical Biology
- MATH 631-632 Introduction to Algebraic Geometry I-II
- MATH 637 Lie Groups
- MATH 776 Topics in Algebraic Number Theory

At the University of Chicago

- MATH 38511 Brownian Motion and Stochastic Calculus, Grade: A-
- MATH 36206 Algebraic Number Theory, Grade: A
- MATH 25700-25800 Honors Basic Algebra I-II
- MATH 20250 Abstract Linear Algebra, Grade: A
- STAT 38300 Measure-Theoretic Probability II, Grade A

TEACHING EXPERIENCE

At the University of Michigan

- Research Experience for Undergraduates (REU): transfer systems — Summer 2026
- Directed Reading Program (DRP): differential Galois theory — Summer 2025
- MATH 115 Calculus I ◇ Instructor — Autumn 2024, Autumn 2025
- MATH 116 Calculus II ◇ Instructor — Winter 2025, Winter 2026, Fall 2026

At the University of Chicago

- MATH 24400 Introduction to Algebraic Geometry ◇ Reader — Winter 2022
- MATH 25400 Basic Algebra II ◇ Super reader — Autumn 2022, Winter 2022
- MATH 25500 Basic Algebra III ◇ Reader — Spring 2022

Note: readers and super readers are teaching assistant positions.

MISCELLANEOUS

Music: I am a classically trained pianist and organist, and I perform regularly. Selected performances:

- Bach: Prelude and fugue in C minor, BWV 546. YouTube
- Bach: Prelude and fugue in E major, BWV 878. YouTube
- Franck: Chorale No. 3 in A minor. YouTube
- Medtner: Sonata-Reminiscenza from Forgotten Melodies, Op. 38 No. 1. YouTube

Languages: Native fluency in Mandarin and English. Learning German.

Citizenship: China.

Date of Birth: December 22, 2001.